

Here is your tentative to-do list for Week 2 to keep you on track with the Challenge.

This week, we recommend you do the following:

1. Finalise your robot's **drive train** type and arrangement.
2. **Manufacture** your robot's prototype (and iterate on it if you discover issues).
3. Make sure all **mechanical components** ('reviving' and 'planting' mechanisms, toggle button kill switch, battery compartment) are installed securely and can operate as intended (check this manually).
4. **Install infrared and RFID sensors** on the robot. Make sure it can climb ramps without damaging the sensors.
5. Install **distance sensors** (front, side(s)) on your robot.
6. Install the Giga+motor shield stack. Connect all cables to your motor shield and tidy up your connection. Consider using **flat ribbon cables for cable management**.
7. Solder two wires to a female **XT30 connector** that will go into your power supply/battery in the future.
8. Program your **mechanical kill switch**. At this stage, for practise you can make it toggle between different operation modes: full forward, full backward, turn left, turn right, stop.

Additional items to practise (ahead of the track and looking forward to Week 3):

1. Wire the **'Reviving' mechanism** to an LED and check its operation.
2. Check **'planter's' operation** when its actuator is powered. You can make it deposit a seed when an RFID tag is detected.
3. Check **distance sensor(s) operation** and logic, e.g. program your robot to stop if something approaches the sensor too closely (can be another robot or a door).
4. Start implementing **basic line following logic** at low speed. You can start from basic controller (bang-bang) and work your way toward PID, if needed. Consider using additional infrared sensors on sides for stability, if needed (although this might not be necessary).
5. Start implementing basic **wall following logic** at low speed.
6. Program the full **'reviving' algorithm**: Make your robot approach an object in front of it carefully until the button is pressed.
7. If you have a dedicated sensor to **detect the 'seed' in the hole**: check its reliability in different scenarios. Adjust code/sensor position for better results.

Needless to say, if something critical (e.g. your drive train) doesn't work as expected, modify it right away without proceeding too far with your coding.

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[\[MEng_RAI\] 2026 Year 1 Robotics Challenge - Trial Run 1 Checklist](#)